

Flood Risk & Mitigation

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FLOOD RISK

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This presentation examines flood risk preparedness and emergency planning through environmental monitoring data and ethical analysis. Each section communicates our data framework, findings, and policy recommendations in a structured, evidence-based sequence.

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DATA SCIENCE EFFORTS



The System

The proposed system consists of two integrated functional components: a flood-risk detection layer and an alert equity assessment layer.

For each watershed included in the 16,200 USGS streamgages, the system computes a Monitoring Density Index (MDI), a metric designed to quantify the extent of sensor coverage relative to watershed drainage area. Watersheds characterized by low MDI values are identified as high-certainty regions, indicating areas where sparse monitoring infrastructure may reduce predictive reliability. In other words, the framework explicitly incorporates and communicates uncertainty within its risk assessments rather than presenting uniformly confident outputs.

The alert equity assessment layer integrates American Community Survey (ACS) census tract data with the flood risk outputs to evaluate disparities in warning accessibility and emergency communication reachability. By combining environmental risk indicators with demographic and infrastructural variables, the system identifies communities that simultaneously exhibit elevated flood vulnerability and limited access to conventional alert dissemination channels. This component is intended to support more equitable disaster preparedness and response strategies by highlighting populations that may be underserved by existing warning systems.

Methods & Outputs

The primary predicting modeling framework employs XGBoost classifiers trained on engineered time-series features derived from hydrological gauge observations. Feature engineering includes

- statistical aggregates
- discharge rate-of-change calculations
- lag-window representations

that are designed to capture short-term temporal dynamics associated with flood onset conditions. Furthermore, regression-based exploratory analysis were conducted using a synthetic disaster-response dataset to examine potential relationships between variables such as emergency response time and recovery duration. As so, they are intended to be interpreted as a supporting broader analysis of disaster-response dynamics.

The system generates three principle outputs intended to support engineering analysis:

1. **Watershed-Level Monitoring Density Index (MDI):** A spatial reliability metric that estimates the adequacy of sensor coverage within each watershed relative to drainage area
2. **Flood Risk Scores for Monitored Locations:** Predictive risk estimates derived from current/historical streamgage conditions to represent the likelihood of flood onset at monitored sites
3. **Reachability Gap Scores by Census Tract:** Composite indicators of flood risk estimates and demographic/infrastructural vulnerability to identify communities with reduced access

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RETENTION PLANNING

The flood-risk prediction system should retain historical environmental and infrastructure data because flood behavior is inherently dependent on long-term pattern recognition across seasons, infrastructure changes, storm frequency, and historical disaster response outcomes. Unlike many short-term machine learning systems, flood-risk forecasting benefits from retaining older data as extreme events may only occur infrequently. As so, removing historical data such as evolution of rainfall intensity, river and stream gauge measurements, soil saturation patterns, and evacuation outcomes could weaken the model's ability to effectively recognize dangerous but rare conditions. Therefore, the project supports a controlled long-term retention strategy rather than unlimited or minimal data retention.





Temporal Drift & Outdated Assumptions

Older environmental data will eventually no longer accurate reflect current flood conditions due to factors such as

- climate change
- urban development
- infrastructure deterioration
- drainage modifications

The model may underestimate emerging risks, overestimate safety, or misallocate emergency resources if it treats outdated data as plausible. To mitigate this issue, the system should undergo periodic recalibration and clearly label historical records with timestamps and confidence indicators.

Overgeneralization Across Regions

When retained data and trained models applied outside of the geographic conditions for which they were originally developed, it may produce unreliable predictions without proper validation. Flood behavior differs substantially across regions due to variations in

- terrain
- infrastructure
- rainfall intensity
- drainage systems

Therefore, retained datasets should include clear geographic metadata and region-specific limitations to prevent inappropriate reuse of the model in unfamiliar contexts.

Infrastructure & Emergency Data Security

Long-term retention of environmental and emergency response data may also create security and operational concerns if sensitive information is improperly stored or exposed over time. In particular, even when datasets do not contain personal information, but are otherwise accessed maliciously, ethical retention practices must require

- secure storage systems
- data versioning
- ongoing monitoring

to ensure that retained information remains both protected and responsibly maintained.

ETHICAL RETENTION RECOMMENDATIONS



The project recommends a controlled long-term retention strategy to ensure that historical flood and infrastructure data remains both useful and ethically managed over time. Raw environmental data should be retained for long-term historical analysis as rare flood events may occur infrequently and older records remain valuable for identifying long-term environmental patterns. However, old infrastructure and emergency-response information should be archived separately and clearly labeled with timestamps, confidence indicators, and geographic metadata to reduce the risk of outdated or regionally inappropriate use. Furthermore, the model should undergo periodic reevaluation following major flood events, infrastructure modifications, or significant climate pattern changes, as historical relationships may no longer accurately reflect current flood conditions over time. Engineering teams should also maintain data version histories, audit trails, and update documentation to ensure transparency regarding what data trained the model, when it was collected, and under which environmental assumptions it was developed. Ultimately, the ethical responsibility of flood-risk systems is not simply whether data is retained, but whether retained information remains contextually accurate, regionally appropriate, secure, and transparently maintained through the model lifecycle.



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ETHICAL REVISIONS

01 Monitoring the Need for Model Calibration

Flood-risk patterns typically shift over time due to climate change, urban development, population growth, drainage modifications, or updating flood-control infrastructure. As so, if the model continues to rely on outdated environmental relationships or traditional monitoring distributions, its predictions may become potentially harmful. In other words, the lack of periodic updates to ensure predictions remain representative of current environmental realities could lead to under-preparation, delayed emergency response, and disproportionate harm to vulnerable populations.

02 Systematic Bias & Unequal Predictive Performance Across Communities

Monitoring infrastructure and data availability are unevenly distributed, the model may produce more reliable predictions in well-monitoring urban regions compared to in rural, low-income, or underserved areas. If audits reveal that certain specific populations consistently experience higher false negative rates, delayed warnings, or reduce prediction quality, the models must be revised to address such disparities. Ethical model maintenance therefore requires ongoing fairness evaluations, incorporation of additional regional data where possible, and adjustments to modeling assumptions or resource-allocation strategies to reduce unequal impacts across communities.

03 Warning Fatigue & Declining Public Trust in Emergency Alerts

Given that the model is designed to prioritize detecting potential flood events, vulnerable populations may receive repeated false warnings for floods, and thus, contributing to a “warning fatigue”, where residents begin to distrust emergency alerts. More specifically, frequent false alarms can impose financial, emotional, and logistical burdens, particularly for lower-income populations who may face such extremities during unnecessary evacuations. If evidence suggests that communities are becoming less response to evacuation recommendations, we are required to update the system to improve calibration, incorporate additional contextual data, or refine warning thresholds to ensure that emergency alerts remain both credible and effective.

MONITORING PROPOSAL

01 Monitoring the Need for Model Calibration

Infrastructure systems and environmental conditions are inherently dynamic due to both natural environmental processes and human-driven interventions. Changes in hydrological patterns, land use, infrastructure availability, or monitoring coverage may substantially alter the statistical relationships on which the model depends. However, the modeling framework should be updated accordingly to ensure that predictions remain representative of current conditions.

02 Systematic Bias & Unequal Predictive Performance Across Communities

Systematic bias is a well-documented risk in socio-technical and data-driven decision systems, particularly when underlying data availability varies across geographic and demographic groups. Given the model's reliance on multi-source observational inputs, disparities in monitoring infrastructure can induce heterogenous predictive performance across regions. To address this, we propose a structured post-deployment auditing framework that evaluates model performance across stratified dimensions, including geographic regions, socioeconomic indicators, and levels of infrastructure access. These audits should incorporate disaggregated error analysis and fairness-oriented performance metrics to identify regions exhibiting persistent performance degradation or elevated uncertainty for early detection of distributional inequities.

03 Warning Fatigue & Declining Public Trust in Emergency Alerts

The potential for false-positive alerts in our early-warning system may lead to repeated non-event warnings. Over time, such inaccuracies can contribute to "alert fatigue", such that affected populations begin to disregard official warnings, reducing the overall effectiveness of the system. To assess this risk, a post-deployment monitoring framework that tracks spatial and temporal patterns of false-positive alerts, complemented by behavioral proxies of engagement, such as evacuation compliance rates or responsiveness to issued warnings can be helpful. However, structured community feedback mechanisms, including periodic surveys, can also be supplemented to assessed perceived credit credibility, trust in the system, and potential behavioral desensitization over time.



01 Monitoring the Need for Model Calibration

A structured program of routine model auditing and performance evaluation is essential for identifying emerging limitations, distributional drift, and potential degradation in predictive accuracy.

1. USGS stream gauge data are collected on an approximately hourly basis, producing a continuously evolving stream of environmental observations.
    a. Full predictive pipeline at hourly intervals is computationally expensive; a more feasible approach would involve conditional or threshold-based model updates
2. In cases where monitoring infrastructure becomes unavailable, system should automatically flag such regions as high-certainty zones and temporarily exclude outdated or incomplete measurements.
    a. Users should be explicitly informed to preserve transparency and prevent overconfidence in model outputs.
    b. Additionally, efforts should be directed toward identifying substitute data sources, proxy variables, or alternative monitoring strategies that can mitigate information gaps created by infrastructure loss.

02 Systematic Bias & Unequal Predictive Performance Across Communities

Identification of regions exhibiting systematically degraded performance should result in implemented targeted mitigation strategies such as deployment of additional sensing infrastructure in under-monitored areas.

1. Model calibration procedures should be conducted to incorporate newly acquired data.
2. Assess whether these interventions lead to measurable improvements in predictive performance.
    a. Re-examination of feature selection, training data representatives, and underlying structural assumptions.



PLAN-OF-RESPONSE

03 Warning Fatigue & Declining Public Trust in Emergency Alerts

If monitoring and community feedback indicate declining trust or evidence of alert fatigue, a set of corrective interventions should be implemented.

1. The model should be evaluated to determine whether internal calibration issues, threshold selection, or feature instability are contributing to elevated false-positive rates.
    a. This can include calibrating decision thresholds to better balance sensitivity and precision, with the goal or reducing low-confidence alerts.
2. Additionally, a region-specific retraining can be necessary to improve local performance characteristics.
    a. If these measures fail to meaningfully improve trust, temporary suspension on automated alerts in the affected region should be considered, with a transition toward human-in-the-loop decision support during the interim period.
3. Finally, transparent communication of the system's limitations, the nature of ongoing corrective actions, and the rationale for any operational changes can help rebuild trust and ensure that stakeholders and affected communities remain informed participants in the adaptive refinement of the system.

SCALING THE SYSTEM



Scaling Risks

Model drift and performance degradation require a defined governance and response framework to ensure that the system remains reliable, equitable, and ethically deployable over time. Corrective actions may include:

- incremental model calibration
- retraining on updated data
- post-deployment monitoring to evaluate the effectiveness of such adjustments.

However, when performance disparities emerge across geographic or demographic groups, more substantial intervention becomes necessary.

Severe Failures

High rates of false positives or negatives also constitute critical operational and ethical concerns. In these situations, automated model outputs should be investigated by human experts and emergency management personnel. A comprehensive documentation process should be implemented to maintain model transparency and accountability. This documentation should record:

- Records of Detected Failures
- Corrective Interventions
- Calibration Procedures
- Subsequent Performance Evaluations

Within high-risk socio-technical systems, a successful model relies on its capacity for rapid detection, responsible mitigation, and adaptive refinement while maintaining public safety and institutional trust.



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FUTURE CONSIDERATIONS

As our flood risk and alert reachability system moves into long-term deployment, engineering teams will need to navigate four broad categories of constraint: technical, legal, financial, and ethical. In particular, ethical considerations will ultimately be the most restricting force, as other constraints largely derive their urgency from ethical obligations rather than existing independently of them.

Technical

Technical issues are considered manageable as model drift due to environmental change, infrastructure updates, and shifting demographic patterns will require periodic revalidation, as previously mentioned. The uneven distribution of USGS streamgage coverage indicates geographic variation in prediction quality, requiring engineering teams to maintain clear versioning and audit trails for documentation transparency. However, adequate institutional commitment, accessible resources, and established practices such as scheduled retraining and confidence labeling will address most standard concerns associated with machine learning deployment.

Legal

Legal exposure for this system centers around the use of census-derived socioeconomic variables as proxies for alert reachability and procedural accountability that contribute to preventable harm during flood events. Although the system is initially designed to serve vulnerable communities, its outputs can later be repurposed by insurance companies, real estate farms, or other stakeholders for exclusionary decision-making, potentially introducing legal and ethical risks under fair housing and anti-discrimination frameworks. Additionally, if public agencies begin relying on the model as an authoritative source of truth, inaccurate risk assessment could contribute to delayed evacuations, unequal emergency response, or insufficient infrastructure preparation in high-risk areas. In these cases, questions of governmental negligence and liability may emerge if communities are harmed after agencies ignored local expertise or failed to communicate the uncertainty and limitations of the model.

Financial

Budget constraints present a significant long-term challenge for sustaining a flood-risk prediction system at scale. While initial deployment costs are substantial, the greater concern is the gradual erosion of critical maintenance practices such as streamgage data integration, model revalidation cycles, multilingual alert infrastructure, and human-in-the-loop oversight. Under financial pressure, these safeguards are often deprioritized as cost-saving measures, despite being essential for maintaining reliability and equity over time. As a result, financial limitations become dangerous through gradual accumulation of technical drift and ethical vulnerabilities.

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FUTURE CONSIDERATIONS

Ethical

The central argument underlying this project is that the flood-risk prediction system was designed to address a problem of environmental justice, rather than solely exploring predictive accuracy. From its inception, the project framed flood response as an issue of equitable alert delivery and resource accessibility for underserved communities in place of simply optimizing forecasting performance. Consequently, the system's effectiveness cannot be evaluated only through aggregate technical metrics. A model that demonstrates strong overall accuracy while consistently producing higher false negative rates in low-income, rural, or under-monitored communities would fail to satisfy the ethical objectives that justified its development.

This distinction has important implications for future engineering and deployment decisions. Factors including

- technical modifications
- legal compliance strategies
- budgetary allocations

must be assessed in terms of whether they preserve or undermine the equity commitments embedded within the original system design. Similarly, expanding the system into new geographic regions without validating performance across different social and infrastructural contents would represent an ethically irresponsible deployment strategy, even if technically feasible.

Ultimately, ethical considerations constitute the most significant constraint on the long-term deployment as they establish the normative framework through which all other decisions must be evaluated.



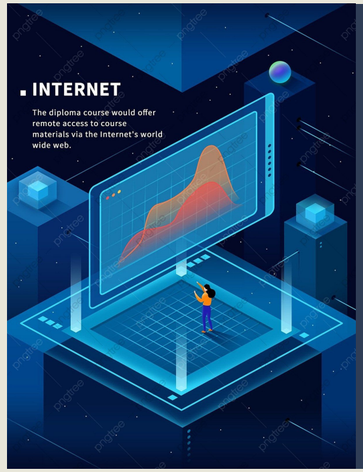
OUR TEAM



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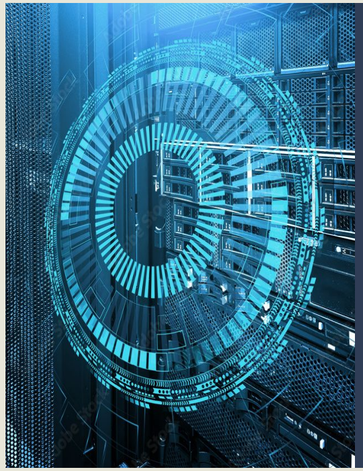
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THANK YOU